Single-Peakedness and the Median Voter

CCSW Ch 10, with Ch 10* for the general version. This is the first *possibility* result we have had in the second half of the course, so it is worth being clear about what exactly is being made possible, and at what price.

The escape being attempted

Arrow's theorem needs **U**, Unrestricted Domain: the rule must work for every logically possible profile. Rejecting U means saying that some profiles will not come up, and that a rule only has to cope with the ones that do.

The question is then whether some *interesting* restriction of the domain lets majority rule behave. Black's answer, from 1948, is yes.

Single-peakedness

Suppose the alternatives can be laid out along a line. Write $a < b$ for "a is to the left of b" on that line. This ordering is a feature of the alternatives, not of anyone's preferences.

R_i is **single-peaked** with respect to $<$ if there is some p_i, person i's **peak**, such that for all a and b:

- if $b < a \leq p_i$, then aP_ib; and
- if $p_i \leq a < b$, then aP_ib.

Moving towards your peak always makes things strictly better. Drawn with the line on the x-axis and preference on the y-axis, each person's preference is a curve that rises to a single summit and falls away from it on both sides.

The standard reading of the line is ideological, left to right. It could also be a tax rate, a date, a budget total, or a speed limit. What matters is that there is one dimension, and that everyone agrees on what it is.

The theorem

> **Note**
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> **Median Voter Theorem (Black 1948).** Let n be odd and let every voter's preference be single-peaked with respect to the same ordering $<$. Let m be the median of the voters' peaks. Then m beats every other alternative in a pairwise majority vote.

Sort the peaks $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(n)}$ and let $m = p_{((n+1)/2)}$, the middle one.

Proof. Take any $a < m$, and let $S = \{i : p_i \geq m\}$. Since m is the $(n+1)/2$-th peak in order, $|S| \geq (n+1)/2$, a strict majority. For each $i \in S$ we have $a < m \leq p_i$, so by single-peakedness mP_ia. So a strict majority prefer m to a, and m beats a.

The case $a > m$ is the mirror image, using $\{i : p_i \leq m\}$. ■

Five lines, and no cleverness. Compare the length of the Arrow proof, and notice that the whole of the work is being done by the domain restriction.

A worked case

Three council members voting on a tax rate, options 0, 10, 20, 35, 50. Peaks at 0, 20 and 50. The median peak is 20.

- 20 against 0: members two and three prefer 20. Carries.
- 20 against 50: members one and two prefer 20. Carries.
- 20 against 10 and against 35: same pattern.

Member two gets their first choice, and neither of the others can build a majority against it. This is the result Hotelling (1929) and Downs (1957) turned into the convergence-to-the-centre argument about two-party competition, which is a game-theoretic argument of the sort we did in the first half.

Why the Condorcet profile is excluded

Back to the three voters from Class 15: $x \succ y \succ z$, $y \succ z \succ x$, $z \succ x \succ y$.

Claim. Single-peakedness rules out the middle alternative of a triple being anyone's *worst*.

Take $x < y < z$ and suppose y is worst for i, so xP_iy and zP_iy. Where is p_i? If $p_i \leq y$, then $p_i \leq y < z$ gives yP_iz, contradicting zP_iy. If $p_i \geq y$, then $x < y \leq p_i$ gives yP_ix, contradicting xP_iy. So there is nowhere for the peak to be.

Now check the Condorcet profile. z is worst for voter one, x is worst for voter two, and y is worst for voter three. Every alternative is somebody's worst. So whichever of the three you try to put in the middle of the line, some voter has it at the bottom, and the profile is not single-peaked under any ordering of the alternatives.

That is the whole mechanism. Single-peakedness is a constraint on which profiles can arise, and the cyclic profile is one of the ones it forbids.

Sen's generalisation

Black's condition is stronger than it needs to be. Sen's own result (Sen 1966, and Ch 10*) isolates what is doing the work.

A profile satisfies **value restriction** over a triple if there is some alternative in the triple, and some rank, such that no voter assigns that alternative that rank. Three ways to satisfy it: some alternative is never best, or never worst, or never in the middle.

> **Note**
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> **Theorem (Sen 1966).** If every triple satisfies value restriction, and the number of voters is odd with no individual indifference, then majority rule yields a strong ordering.

Single-peakedness gives "some alternative is never worst", as we just showed, so it is one special case. "Never best" is the single-caved case, where preferences dip in the middle instead of peaking. "Never medium" has no natural picture at all, which is part of Sen's point: the condition that matters is combinatorial, and the geometric story about a left-right line is one illustration of it rather than the thing itself.

The Condorcet profile fails all three forms. Every alternative is best for someone, worst for someone, and in the middle for someone. It is as far from value-restricted as a three-voter profile can be.

Four things to say against this escape

The dimension is extra information. Single-peakedness is not a property of a profile. It is a property of a profile *relative to an ordering of the alternatives*, and that ordering is not contained in anyone's preferences. So the escape works by bringing in structure from outside the preference data, which is the same move as the cardinality and interpersonal-comparison escapes in Ch 7, dressed differently. Whoever gets to say what the dimension is has made part of the decision.

It fails in two dimensions. Restrict to one dimension and you get a Condorcet winner. Allow two, and McKelvey (1976) showed that majority rule cycles can wander anywhere in the space: for any two points there is a finite sequence of majority votes leading from one to the other. Plott's (1967) conditions for a majority core in several dimensions are knife-edge and will not hold generically. One dimension is not a mild idealisation of two.

It does not save Arrow's theorem. It saves majority rule on a restricted domain. The theorem is untouched, since it was a theorem about rules defined on the unrestricted domain. What we have learned is where to look for rules that work, not that Arrow was wrong.

Whether real preferences are single-peaked is an empirical question, and the answer is domain-dependent. Legislative voting on a budget total, plausibly yes. Preferences over candidates who differ on several issues at once, much less so. The condition is checkable, which is a virtue, and it is often false.

Exercises

1. Verify directly that the Condorcet profile is not single-peaked under any of the orderings $x < y < z$, $x < z < y$, $y < x < z$. Why are the other three orderings already covered?
2. With an even number of voters, the median is not unique. Construct a four-voter single-peaked profile where two alternatives tie under majority rule. Is there still no cycle?

3. Show that if everyone's preference is single-peaked and all the peaks are distinct, the median voter's peak is also the outcome of the rule "take the alternative that beats most others head to head".

4. Give a profile over three alternatives that is value-restricted but not single-peaked under any ordering. (Try "never best".)

5. Single-peakedness constrains the *profile*. Property α and property β from Ch 1* constrain the *choice function*. Are these two kinds of constraint doing the same job?